

Optimism Under Uncertainty: Regret Analysis of the Upper Confidence Bound Algorithm for Multi-Armed Bandits

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Abstract

The stochastic multi-armed bandit problem arises naturally in clinical trials, A/B testing, and recommendation systems, where a decision-maker must learn which option is best while simultaneously using that knowledge to maximize cumulative reward. This paper formally defines the multi-armed bandit problem, policies, and the regret metric. We present a naive baseline policy, Explore-Then-Commit (ETC), for comparison. Then, we introduce the classical Upper Confidence Bound (UCB1) algorithm and provide a clear and self-contained proof of its regret upper bound, showing that the bound grows logarithmically with time. A simple example in advertisement optimization demonstrates that UCB1 achieves substantially lower regret upper bound than the ETC baseline.

1 Introduction

Imagine a gambler facing a slot machine with multiple levers, each producing rewards drawn from a different and unknown distribution. The player can pull the levers only a limited number of times—for instance, 1,000. How should they choose among these arms to maximize their total accumulated reward (see Fig. 1, created with Google Nano-Banana)? This setting illustrates the well-known *multi-armed bandit* (MAB) problem.

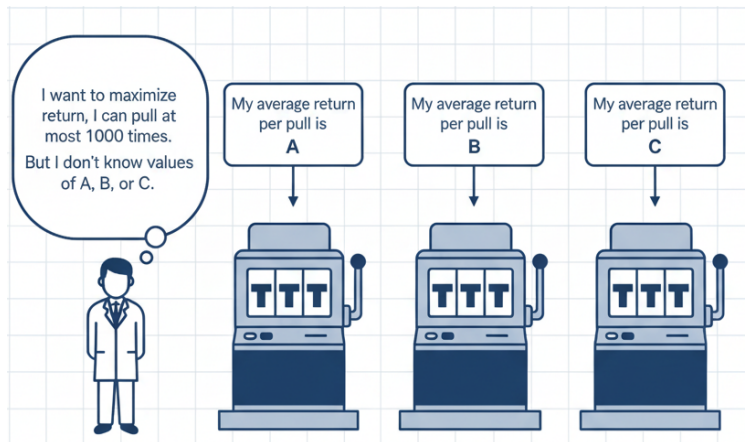


Figure 1: Illustration of the Multi-Armed Bandits Problem

The origins of the MAB problems trace back to early work in clinical trials, where researchers sought to balance *learning* and *earning*. Suppose there are two treatments—each analogous to an arm of the slot machine—with unknown success probabilities. As we gather evidence to determine which treatment is superior, we simultaneously wish to assign more patients to the better treatment, thereby improving the overall recovery rate [7]. This trade-off between *exploration* (e.g. gathering more data to draw stronger conclusion) and *exploitation* (e.g. utilizing existing data to try to cure more patients) is the key for the MAB problem [6].

Over time, the MAB problem and its *algorithms* have been extended to many fields. They now play a central role in A/B testing, where one must decide which product version performs better, and in recommendation systems, where different content categories act as arms and user engagement (e.g., click-through rate) serves as the reward signal. Bandit algorithms are also foundational in reinforcement learning, where agents repeatedly make decisions under uncertainty [6].

In the setting of MAB, the performance of an algorithm is typically measured by its *regret*, which represents the difference between the maximum possible reward an agent could have obtained and the reward it actually receives. For example, in the scenario of Figure 1, suppose that A is the highest among the three average returns of the machines. Then, the “gambler’s regret” is the result of the times the gambler did not pull A. In the context of clinical trials, a lower regret corresponds to fewer patients who do not receive optimal treatment, which illustrates the importance of reducing the regret. In short, minimizing regret is equivalent to maximizing the cumulative reward, and provides a principled way to quantify losses of strategies [6].

A wide variety of algorithms have been developed to address the MAB problem, including greedy, ϵ -greedy, Thompson sampling, and MOSS [6, 2, 1]. Among them, the *Upper Confidence Bound (UCB1)* algorithm stands out as one of the most classical and influential approaches. It is simple, parameter-free, and supported by clear regret bounds [6, 2]. Moreover, its principle of *optimism under uncertainty* has inspired a broad family of algorithms such as Monte Carlo Tree Search [5]. Therefore, we are interested in how UCB1 reduced the regrets by balancing exploration and exploitation, and how we can bound its regrets.

In the rest of the paper, we will formally define the stochastic multi-armed bandit problem in Section 2, including the basic components such as arms, actions, rewards, and the overall objective of maximizing cumulative reward. In Section 3, we introduce key mathematical definitions including optimal arms, sub-optimality gaps, histories, policies, and the regret metric that measures the performance of a policy. To illustrate these concepts, Section 4 presents the *Explore-Then-Commit (ETC)* policy as a naive baseline policy, demonstrating the fundamental exploration-exploitation dilemma and motivating the need for more adaptive approaches. In Section 5, we present the Upper Confidence Bound (UCB1) policy, explain the principle of optimism under uncertainty, and derive its confidence radius based on Chernoff–Hoeffding bound. Finally, in Section 6, we provide a complete proof of the regret bounds for UCB1.

2 Multi-Armed Bandits

We will formally define the Multi-Armed Bandit (also called K-Armed Bandit, or MAB), present its related concepts, and explain its goal in this section.

Definition 2.1 (Multi-Armed Bandit). A *multi-armed stochastic bandit* consists of K arms indexed by $i \in \mathcal{A} = \{1, 2, \dots, K\}$. Each arm i has an unknown reward distribution P_i with support in $[0, 1]$ and expected reward $\mu_i = \mathbb{E}_{X \sim P_i}[X]$. A *learner* (also called a *player* or *agent* in different literatures) sequentially pulls these arms in order to maximize the cumulative reward.

The definition of MAB naturally leads to two concepts, the first of which is action.

Definition 2.2 (Action and Pull Count). An *action* at time t is the arm selected, denoted $I_t \in \mathcal{A}$ where $\mathcal{A}$ is called *action space*. The *pull count* of arm i up to time t is

$$T_i(t) = \sum_{s=1}^t \{I_s = i\},$$

where $\{\cdot\}$ is the indicator function. In particular, the pull count of the optimal arm is also denoted by $T^*(t)$.

Another concept is reward. Intuitively, it is the result or feedback from an action.

Definition 2.3. (Reward) When arm i is pulled for the s -th time, a *reward* $X_{i,s}$ is sampled independently from a distribution P_i . We use two equivalent notations:

- X_t : the reward observed at time t (time-indexed)

- $X_{i,s}$: the s -th reward from arm i (arm- and pull-indexed)

These relate via: if $I_t = i$ and $T_i(t-1) = s-1$, then $X_t = X_{i,s}$ and $T_i(t) = s$.

Note that we assume the support of P_i for all i is $[0, 1]$ in this paper.

Example 2.4 (Advertisement Selection as a MAB Problem). Suppose you are an advertisement manager designing two versions of an ad for a new product, “XPhone,” to be displayed on YouTube. Let the two ads be indexed by $i \in \{1, 2\}$. The *purchasing rate* (i.e., probability that a viewer purchases the product after seeing ad i) of the two ads are parameters p_1 and p_2 respectively, which are unknown. In other words, for each ad i , the indicator of a purchase $\{\text{purchase}\}$ follows a Bernoulli distribution $\text{Bern}(p_i)$.

You have a budget that allows exactly 1000 ad displays, and you wish to maximize the total number of purchases (equivalently, the total reward).

In this setup:

- The two *arms* correspond to the two ad versions.
- The *reward distributions* are Bernoulli with unknown means p_1 and p_2 .
- At each round t , the *action* is to choose which version $I_t \in \{1, 2\}$ to display.
- If you display advertisement 2 at round $t = 10$, then $I_{10} = 2$.
- If advertisement 2 has been displayed 6 times up to round 9, then its *pull count* at $t = 9$ is $T_2(9) = 6$.
- The observed reward X_t is 1 if the viewer buys “XPhone” and 0 otherwise.
 - If the viewer at round 10 makes a purchase, then $X_{10} = 1$.
 - If, at round 10, arm 2 is being pulled for the 7-th time ($T_2(10) = 7$), then we also have $X_{2,7} = 1$.

Remark 2.5. This example illustrates the exploration–exploitation dilemma: should the manager allocate more displays to the ad that currently performs better, or explore the other ad to obtain a more accurate estimate of its success probability?

The example above illustrates the **goal of the multi-armed bandit problem**—to maximize the cumulative reward $\sum_{t=1}^T X_t$. Since each X_t is a random variable, we are typically interested in maximizing its expected value, that is,

$$\mathbb{E} \left[\sum_{t=1}^T X_t \right]. \quad (1)$$

3 Policies

Example 2.4 in Section 2 highlights why we need to introduce the concept of a *policy* (also called an *allocation strategy* or *bandit algorithm*). In that scenario, the advertisement manager faces two challenges. First, as discussed in Remark 2.5, they must balance exploration and exploitation—deciding how to allocate the display budget between the two ad versions to maximize the number of purchases. This allocation strategy is precisely what we will formalize as a *policy*. Second, the manager must make repeated decisions, potentially thousands or even millions of times, such as choosing which ad to show for each of the 1000 displays. It would be infeasible to make such decisions manually for every round. Therefore, this strategy must operate automatically, mapping past observations to future actions.

Before introducing the formal definition of a policy, we first define the concept of a *history*.

Definition 3.1 (History). A *history* $H_t \in \mathcal{H}$ at time t is all past information up to time t :

$$H_t = (I_1, X_1, I_2, X_2, \dots, I_t, X_t),$$

where the *history space* is defined as $\mathcal{H} := \bigcup_{t=0}^{\infty} (\mathcal{A} \times [0, 1])^t$. Note that $H_0 := \emptyset$.

Before deciding which arm to pull at round t , the learner has access to all previously chosen actions and the corresponding rewards H_{t-1} . Naturally, the learner would like to utilize this accumulated information to make increasingly better decisions over time. Therefore, a *policy* can be defined as a map from histories to actions.

Definition 3.2 (Policy). A *policy* is a mapping $\pi : \mathcal{H} \rightarrow \mathcal{A}$ from histories to actions, where $\mathcal{H}$ is the set of all possible histories. At time t , the policy selects $I_t = \pi(H_{t-1})$ based on past observations.

Remark 3.3. The *policy* in Definition 3.2 is actually *deterministic policy*. There is also *stochastic policy* which we will not discuss in this paper.

While the practical goal of the learner is to maximize total reward, this absolute quantity depends on the unknown means μ_i and is difficult to compare across algorithms. As we briefly mentioned in Section 1, we usually use *regret*—the expected loss compared to always pulling the best arm—as a normalized performance measure to evaluate and compare a policies. We introduce the notation of optimal arm, optimal mean, and suboptimal gap; and then define the regrets.

Definition 3.4 (Optimal Arm and Optimal Mean). The *optimal arm* is $i^* = \arg \max_{i \in \mathcal{A}} \mu_i$ with *optimal mean* $\mu^* = \mu_{i^*}$. We assume uniqueness of i^* for simplicity (ties can be broken arbitrarily).

We first define the difference between the expected reward of a specific arm and that of the optimal arm.

Definition 3.5 (Suboptimal Gap). The *suboptimality gap* of arm i is $\Delta_i = \mu^* - \mu_i$.

While the suboptimal gap measures the loss in a single step, we are primarily interested in the cumulative loss over time. This is captured by the random regret, which compares the learner’s performance against an *oracle policy* (i.e., a policy that always selects the optimal arm).

Definition 3.6 (Random Regret). The *random regret* up to time n is

$$R_n = n\mu^* - \sum_{t=1}^n X_t.$$

As mentioned in Section 1, because X_t is a random variable, and we are interested in maximizing the expectation of cumulative reward (1), we take the expected value of the random regret. The objective of policies is to minimize their expected regret.

Definition 3.7 (Expected Regret). The *expected regret* of a policy is expected value of the random regret: $\mathbb{E}[R_n] = n\mu^* - \mathbb{E}[\sum_{t=1}^n X_t]$.

Next, we will use a naive policy to explain these concepts.

4 Explore-Then-Commit

In this section, we will present a **naive baseline policy, Explore-Then-Commit (ETC)** adapted from *Bandit Algorithms* [6], to illustrate the definitions in section 3. First, we introduce a useful definition: sample mean (also called empirical mean).

Definition 4.1 (Sample Mean). After pulling arm i a total of s_i times, its *empirical mean* reward is

$$\bar{X}_{i,s_i} = \frac{1}{s_i} \sum_{t=1}^{s_i} X_{i,t}.$$

In particular, the *empirical mean* of the optimal arm is also denoted by $\bar{X}_s^*$ where $s = s_{i^*}$. This is our estimate of the true mean μ_i based on observed samples.

Then, we specify the ETC algorithm [6].

Algorithm 1 ETC Algorithm

- 1: **Input:** Number of arms K , number of exploration on each arm m , time horizon n
 - 2: **Initialization:** Pull each arm m times (for $t = 1, 2, \dots, mK$) and get statistics $\bar{X}_{i,m} \forall i \in \mathcal{A}$.
 - 3: **for** $t = mK + 1, mK + 2, \dots, n$ **do**
 - 4: Play arm i that maximizes $\bar{X}_{i,m}$ (ties can be broken arbitrarily).
 - 5: **end for**
-

We use the setting of Example 2.4 to explain how the Policy 1 works.

Example 4.2. For any s_1, s_2 , let $X_{1,s_1} \sim \text{Bern}(0.6)$ and $X_{2,s_2} \sim \text{Bern}(0.3)$, we use Policy 1 with $m = 2$. Then, the first four *actions* are fixed. So, the *history* up to $t = 4$ is

$$H_4 = \{I_1 = 1, X_1 = X_{1,1}, I_2 = 1, X_2 = X_{1,2}, I_3 = 2, X_3 = X_{2,2}, I_4 = 2, X_4 = X_{2,2}\}.$$

Therefore, the each ad is displayed twice at the first 4 rounds, which we call exploration. We suppose the viewers purchased the “XPhone” in the first three rounds, but not in the 4th round. Then, the algorithm compute the sample mean of each ad:

$$\bar{X}_{1,2} = \frac{1}{2}(X_{1,1} + X_{1,2}) = \frac{1}{2}(1 + 1) = 1 \quad \text{and} \quad \bar{X}_{2,2} = \frac{1}{2}(X_{2,1} + X_{2,2}) = \frac{1}{2}(1 + 0) = 0.5$$

For the rest rounds, based on the Policy 1 we always display the ad with highest sample mean of purchases in its first two displays, which we call exploitation.

$$I_t = \operatorname{argmax}_i \bar{X}_{i,2} = 1 \text{ for } t = 5, \dots, 1000.$$

Suppose in the following 996 rounds, 580 viewers purchased “XPhone”. The *random regret* in this case is

$$R_n = n\mu^* - \sum_{t=1}^{1000} X_t = 1000 \times 0.6 - 583 = 17.$$

By Theorem 6.1 in [6](ETC with $1 \leq m \leq n/K$), the *expected regret*

$$\mathbb{E}[R_n] \leq m \sum_{i=1}^K \Delta_i + (n - mK) \sum_{i=1}^K \Delta_i \exp\left(-\frac{m\Delta_i^2}{4}\right) = 2 \cdot 0.3 + (1000 - 4) \cdot 0.3 \cdot e^{-\frac{2 \cdot (0.3)^2}{4}}.$$

That is,

$$\mathbb{E}[R_n] \leq 0.6 + 996 \cdot 0.3 \cdot e^{-0.045} \approx 286.25. \tag{2}$$

This means that, on average, the ETC policy with $m = 2$ is expected to yield at most 286.25 fewer purchases than an oracle strategy that always selects the optimal advertisement.

See [6] for a detailed analysis and a proof of *expected regret* upper bound of ETC.

It seems that the random regret in the example above is acceptable. However, as we can see, its expected regret is relative high. That is because if the sample mean of the optimal advertisement is underestimated in the first 4 rounds, then the following 996 rounds will be allocated to inferior advertisement, causing high random regret. If we want to stabilize the statistics, we need the explore more times. Then, the non-optimal advertisement would be selected a lot of times and the budget will be wasted in the exploration. This dilemma motivates us to design an adaptive policy that better balance the exploration and exploitation.

5 Upper Confidence Bound

In this section, we will introduce the idea of *optimism under uncertainty* and explain how this idea inspired the design of the *Upper Confidence Bound (UCB1) Policy*. The main idea of *optimism under uncertainty* is: when data about an option are scarce, one should act as if the environment could be as good as plausibly consistent with the data collected so far [6]. First, we define *confidence radius* that measures *uncertainty*.

Definition 5.1 (Confidence Radius). For arm i pulled s_i times at time t , we define the *confidence radius* as

$$c_{t,s_i} = \sqrt{\frac{2 \ln t}{s_i}}.$$

In particular, the confidence radius of the optimal arm is also denoted by $c_{t,s}$ where $s = s_{i^*}$.

Remark 5.2. When the policy is deciding on action to take at time t , it only know c_{t-1,s_i} where $s_i = T_i(t-1)$.

We notice that this radius grows logarithmically with time t and shrinks with the square root of the number of pulls s_i . Intuitively, when an arm i has been pulled fewer times, the *uncertainty* associated with it is higher, resulting in a larger confidence radius. Under such uncertainty, we adopt an *optimistic view*: we assume that the true mean of arm i could be significantly higher than its current sample mean. To formalize this optimism, we assign each arm an Upper Confidence Bound.

Definition 5.3 (Upper Confidence Bound). The *Upper Confidence Bound* of an arm i is the sum of its sample mean and its confidence radius: $\text{UCB}_i(t-1) := \bar{X}_{i,s_i} + c_{t-1,s_i}$ where $s_i = T_i(t-1)$.

The UCB1 algorithm, adapted from Auer et. al.'s paper [2], selects arms based on their Upper Confidence Bound.

Algorithm 2 UCB1 Algorithm

- 1: **Input:** Number of arms K , time horizon n
 - 2: **Initialization:** Pull each arm once (for $t = 1, 2, \dots, K$)
 - 3: **for** $t = K + 1, K + 2, \dots, n$ **do**
 - 4: Play arm j that maximizes $\bar{X}_{j,T_j(t-1)} + c_{t-1,T_j(t-1)}$,
 - 5: Observe reward X_t and update statistics.
 - 6: **end for**
-

This *upper confidence bound* represents an optimistic estimate of its true conversion rate—one that, with high probability, slightly *overestimates* the unknown mean. Intuitively, if the upper confidence bound assigned to the optimal advertisement (say, Ad 1), UCB_1 , is indeed an overestimate, then the inferior ad (Ad 2) can only be chosen if its own upper confidence bound UCB_2 happens to exceed UCB_1 . However, the upper confidence bound of Ad 1 UCB_1 is higher than its true mean μ^* . And each time Ad 2 is shown, we gain more data about it, tightening its confidence interval so that its upper bound will eventually fall below that of Ad 1. Thus, the number of times a suboptimal ad is mistakenly displayed is limited. Unlike the ETC policy that explores for a fixed time and then only takes sample mean into consideration, UCB keeps a balance of exploration and exploitation by adding a confidence radius term and updating sample mean dynamically.

The confidence radius is not a pure empirical choice. To understand the reason why we choose it, we first introduce the Chernoff–Hoeffding bound, which is useful for bounding the sum of random variables. See John Duchi's lecture notes [4] for a complete proof of this lemma.

Lemma 5.4 (Chernoff–Hoeffding bound). Let $X_1, \dots, X_n$ be random variables with support in $[0, 1]$ and such that

$$\mathbb{E}[X_t \mid X_1, \dots, X_{t-1}] = \mu.$$

Let $S_n = X_1 + \dots + X_n$. Then for all $a \geq 0$,

$$\mathbb{P}\{S_n \geq n\mu + a\} \leq e^{-2a^2/n} \quad \text{and} \quad \mathbb{P}\{S_n \leq n\mu - a\} \leq e^{-2a^2/n}.$$

Using the Chernoff–Hoeffding bound, we prove the following lemma, which demonstrates that the confidence radius is designed to bound the deviation of the empirical mean from the true mean and control the probability of estimation error.

Lemma 5.5 (Confidence Interval). *Suppose arm i has been pulled s_i times at time t , then with $c_{t,s_i} = \sqrt{2 \ln t / s_i}$, we have*

$$\mathbb{P} \left\{ \mu_i \geq \bar{X}_{i,s_i} + \sqrt{\frac{2 \ln t}{s_i}} \right\} \leq t^{-4} \quad \text{and} \quad \mathbb{P} \left\{ \mu_i \leq \bar{X}_{i,s_i} - \sqrt{\frac{2 \ln t}{s_i}} \right\} \leq t^{-4}.$$

Proof. For rewards of arm i with support in $[0, 1]$ and at any time t , we take the second inequality 5.4, divide both side of the event inequality in $\mathbb{P}\{\}$ by s_i , and let the RHS be δ . By rearranging, we have

$$\mathbb{P}\{\mu \geq \bar{X}_{i,s_i} + a/s_i\} \leq e^{-2a^2/s_i} = \delta \tag{3}$$

$$-2a^2/s_i = \ln \delta \tag{4}$$

$$a = \sqrt{-s_i \ln \delta / 2} = \sqrt{s_i \ln(1/\delta) / 2} \tag{5}$$

$$\mathbb{P} \left\{ \mu \geq \bar{X}_{i,s_i} + \left(\sqrt{s_i \ln(1/\delta) / 2} \right) / s_i \right\} = \mathbb{P} \left\{ \mu \geq \bar{X}_{i,s_i} + \left(\sqrt{\ln(1/\delta) / (2s_i)} \right) \right\} \leq \delta \tag{6}$$

Setting $\delta := t^{-4}$

$$\sqrt{\frac{\ln(1/t^{-4})}{2s_i}} = \sqrt{\frac{4 \ln t}{2s_i}} = \sqrt{\frac{2 \ln t}{s_i}} \quad \text{and} \quad \mathbb{P} \left\{ \mu \geq \bar{X}_{i,s_i} + \sqrt{\frac{2 \ln t}{s_i}} \right\} \leq t^{-4}. \tag{7}$$

The proof is concluded by plugging in back to (6) and doing the same with the first inequality of Lemma 5.4. $\square$

This choice makes each “bad” event occur with probability $\lesssim t^{-4}$. When we later union-bound over all rounds t and all internal counters $s, s_i \leq t$, the triple sum of these probabilities behaves like $\sum_{t \geq 1} t^2 \cdot t^{-4} = \sum_{t \geq 1} t^{-2}$, a convergent p -series. This is exactly the step that produces the finite constant in our analysis; see inequality (27) in Section 6.

6 Regret Bounds of UCB1

We will prove the regret bounds of UCB1 algorithm in this section. First, we introduce the following lemmas that are necessary for the proof.

According to Definition 3.7, the expected regret is depend on the expected reward $\mathbb{E}[X_t]$ of each pull. Since $\mathbb{E}[X_{i,t}] = \mu_i$ is a constant for each i , we only need to bound $\mathbb{E}[T_i(T)]$ for each i to bound the expected regret. Therefore, we introduce the Lemma 6.1 [6].

Lemma 6.1 (Regret Decomposition). *The expected regret decomposes as*

$$\mathbb{E}[R_T] = \sum_{i=1}^K \Delta_i \mathbb{E}[T_i(T)].$$

The following result, originally solved by Euler in 1734 (see [3]), establishes the value of the Basel series, which we will use in proof later:

Lemma 6.2 (Euler’s Solution to the Basel Problem). *We have*

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

We are going to prove the main theorem, regret upper bound of UCB1. The theorem and the proof is adapted from Auer et. al.'s paper [2].

Theorem 6.3 (Regret Upper Bound of UCB1). *For all $K > 1$, if policy UCB1 is run on K machines having arbitrary reward distributions $P_1, \dots, P_K$ with support in $[0, 1]$, then, after any number n of plays, the upper bound of expected regret $\mathbb{E}(R_n)$ is*

$$\left[8 \sum_{i: \mu_i < \mu^*} \left(\frac{\ln n}{\Delta_i} \right) \right] + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right),$$

where $\mu_1, \dots, \mu_K$ are the expected values of $P_1, \dots, P_K$.

Proof. First, we introduce the following notation. For any predicate Π , we define $\{\Pi(x)\}$ to be the indicator function of the event $\Pi(x)$, then

$$\{\Pi(x)\} = \begin{cases} 1, & \text{if } \Pi(x) \text{ is true,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $c_{t,s} = \sqrt{(2 \ln t)/s}$. For any machine i , we upper bound its pull count $T_i(n)$ on any sequence of plays.

Step 1: Peel off an initial exploration budget. Based on the Algorithm 2, each arm must be pulled once in the first K rounds. So, if $n \leq K$, it is trivial that $T_i(n) \leq 1$. Suppose $n \geq K + 1$, it is obvious that

$$T_i(n) = 1 + \sum_{t=K+1}^n \{I_t = i\}. \quad (8)$$

We expand this strategy and introduce a threshold $\ell \in \mathbb{N}^+$. Then, there are two cases. The first case is trivial: if arm i is pulled no more than ℓ times, inequality (9) is obviously true. Otherwise, arm i must be pulled strictly more than ℓ times. Then, $T_i(n)$ consists of the first ℓ pulls and the pulls after that. After the arm has been pulled ℓ times, $T_i(t-1) \geq \ell$ become true, then the summation term will begin to count the pulls. Therefore, the inequality (9) holds.

$$T_i(n) \leq \ell + \sum_{t=K+1}^n \{I_t = i, T_i(t-1) \geq \ell\} \quad (9)$$

Step 2: Reduce “choose i ” to a single comparison of UCBs with the optimal arm. $I_t = i \iff$ arm i is pulled at time $t \iff$ arm i has the highest UCB value at time $t \implies$ the UCB of arm i is higher than the UCB of optimal arm at time t . The condition is relaxed in this step. Therefore, we have

$$T_i(n) \leq \ell + \sum_{t=K+1}^n \{\bar{X}_{T^*(t-1)}^* + c_{t-1, T^*(t-1)} \leq \bar{X}_{t, T_i(t-1)} + c_{t-1, T_i(t-1)}, T_i(t-1) \geq \ell\} \quad (10)$$

Step 3: Remove the random pull counts via min/max. For the first event of the indicator function in inequality (10), we can continue to relax the condition by taking the minimum of LHS and the maximum of the RHS with respect to $s \in (0, t)$

$$\min_{0 < s < t} \bar{X}_s^* + c_{t-1, s} \leq \bar{X}_{T^*(t-1)}^* + c_{t-1, T^*(t-1)} \quad (11)$$

$$\bar{X}_{t, T_i(t-1)} + c_{t-1, T_i(t-1)} \leq \max_{\ell \leq s_i < t} \bar{X}_{i, s_i} + c_{t-1, s_i}. \quad (12)$$

Since the index range $\ell \leq s_i < t$ implies the condition $T_i(t-1) \geq \ell$, along with the inequalities (11) and (12), we have

$$T_i(n) \leq \ell + \sum_{t=K+1}^n \left\{ \min_{0 < s < t} \bar{X}_s^* + c_{t-1, s} \leq \max_{\ell \leq s_i < t} \bar{X}_{i, s_i} + c_{t-1, s_i} \right\}. \quad (13)$$

Step 4: Expand the “there exists (s, s_i) ” event by a union bound. In inequality (13), the event

$$\min_{0 < s < t} \bar{X}_s^* + c_{t-1,s} \leq \max_{\ell \leq s_i < t} \bar{X}_{i,s_i} + c_{t-1,s_i} \quad (14)$$

implies that there exists a pair (s, s_i) where s and s_i are in their respective range such that the inequality

$$\bar{X}_s^* + c_{t-1,s} \leq \bar{X}_{i,s_i} + c_{t-1,s_i} \quad (15)$$

holds, so the event (14) is equivalent to

$$\bigcup_{s=1}^{t-1} \bigcup_{s_i=\ell}^{t-1} \{ \bar{X}_s^* + c_{t-1,s} \leq \bar{X}_{i,s_i} + c_{t-1,s_i} \}. \quad (16)$$

There might be multiple pairs (s, s_i) such that event (14) holds, so we have

$$\left\{ \min_{0 < s < t} \bar{X}_s^* + c_{t-1,s} \leq \max_{\ell \leq s_i < t} \bar{X}_{i,s_i} + c_{t-1,s_i} \right\} \leq \sum_{s=1}^{t-1} \sum_{s_i=\ell}^{t-1} \{ \bar{X}_s^* + c_{t,s} \leq \bar{X}_{i,s_i} + c_{t,s_i} \}.$$

We also relax the index of the first summation to the range of 1 to ∞ so that we can exploit the result of p-series later. Then, we relax inequality (13) and get

$$T_i(n) \leq \ell + \sum_{t=1}^{\infty} \sum_{s=1}^{t-1} \sum_{s_i=\ell}^{t-1} \{ \bar{X}_s^* + c_{t,s} \leq \bar{X}_{i,s_i} + c_{t,s_i} \}. \quad (17)$$

Step 5: Split the complicated event into three simple events. Now, observe that $\bar{X}_s^* + c_{t,s} \leq \bar{X}_{i,s_i} + c_{t,s_i}$ (event (15)) implies that at least one of the following must hold

$$\bar{X}_s^* \leq \mu^* - c_{t,s} \quad (\text{underestimation of optimal arm}) \quad (18)$$

$$\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i} \quad (\text{overestimation of suboptimal arm}) \quad (19)$$

$$\mu^* < \mu_i + 2c_{t,s_i} \quad (\text{overlapped confidence interval}). \quad (20)$$

This can be proved by contrapositive. If we negate all the three inequalities above, then the inequality (15) will be false.

We bound the probability of events (18) and (19) using Lemma 5.5

$$\mathbb{P}\{\bar{X}_s^* \leq \mu^* - c_{t,s}\} \leq e^{-4 \ln t} = t^{-4}$$

$$\mathbb{P}\{\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i}\} \leq e^{-4 \ln t} = t^{-4}.$$

So,

$$\mathbb{P}((\bar{X}_s^* \leq \mu^* - c_{t,s}) \cup (\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i})) \leq 2t^{-4}. \quad (21)$$

Step 6: Eliminate event (20) by choosing a proper ℓ . Then, we want to guarantee event (20) not happening. We negate (20)

$$\mu^* > \mu_i + 2c_{t,s_i}.$$

Since $c_{t,s_i} = \sqrt{\frac{2 \ln t}{s_i}}$, we have

$$2\sqrt{\frac{2 \ln t}{s_i}} < \mu^* - \mu_i = \Delta_i.$$

By taking square on both side and multiplying by s_i on both side, we get

$$s_i > \frac{8 \ln t}{\Delta_i^2}.$$

So, we let $\ell = \lceil (8 \ln n) / \Delta_i^2 \rceil$, then $s_i \geq \ell \geq (8 \ln n) / \Delta_i^2 \geq (8 \ln t) / \Delta_i^2$. So, (20) is false. In this case, event (15) is contained in the union of events (18) and (19). So, from inequality (17), we have

$$T_i(n) \leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{t=1}^{\infty} \sum_{s=1}^{t-1} \sum_{s_i=\lceil (8 \ln n) / \Delta_i^2 \rceil}^{t-1} \{(\bar{X}_s^* \leq \mu^* - c_{t,s}) \cup (\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i})\}. \quad (22)$$

Step 7: Take expectations and convert indicators to probabilities. We take expected value on both side. By linearity of expectation, we have

$$\mathbb{E}[T_i(n)] \leq \mathbb{E} \left[\left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{t=1}^{\infty} \sum_{s=1}^{t-1} \sum_{s_i=\lceil (8 \ln n) / \Delta_i^2 \rceil}^{t-1} \{(\bar{X}_s^* \leq \mu^* - c_{t,s}) \cup (\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i})\} \right] \quad (23)$$

$$\leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{t=1}^{\infty} \sum_{s=1}^{t-1} \sum_{s_i=\lceil (8 \ln n) / \Delta_i^2 \rceil}^{t-1} \mathbb{E} \{(\bar{X}_s^* \leq \mu^* - c_{t,s}) \cup (\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i})\} \quad (24)$$

$$\leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{t=1}^{\infty} \sum_{s=1}^{t-1} \sum_{s_i=\lceil (8 \ln n) / \Delta_i^2 \rceil}^{t-1} \mathbb{P}((\bar{X}_s^* \leq \mu^* - c_{t,s}) \cup (\bar{X}_{i,s_i} \geq \mu_i + c_{t,s_i})) \quad (25)$$

By inequality (21), we have

$$\leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{t=1}^{\infty} \sum_{s=1}^t \sum_{s_i=1}^t 2t^{-4} \quad (26)$$

Since t does not depend on s or s_i ,

$$\leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{t=1}^{\infty} 2t^{-2} \quad (27)$$

Note that here explains why $\delta = t^{-4}$ is important in Lemma 5.5. Otherwise, the series would be divergent. We eliminate the ceiling sign by adding 1 and use Lemma 6.2. We get

$$\leq \frac{8 \ln n}{\Delta_i^2} + 1 + \frac{\pi^2}{3} \quad (28)$$

Finally, we plug in this result to the Lemma 6.1 and get

$$\mathbb{E}[R_T] \leq \sum_{i=1}^K \Delta_i \left(\frac{8 \ln n}{\Delta_i^2} + 1 + \frac{\pi^2}{3} \right) \quad (29)$$

Note $\Delta_{i^*} = 0$, so we only consider suboptimal arms for the first term.

$$\mathbb{E}[R_T] \leq \left[8 \sum_{i: \mu_i < \mu^*} \frac{\ln n}{\Delta_i} \right] + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right) \quad (30)$$

which concludes the proof. $\square$

We use the setting of Example 4.2 and compute the expected regret upper bound of UCB1 policy with Theorem 6.3

Example 6.4. In our case, $K = 2$, $T = 1000$, $\mu^* = 0.6$, and the gaps are $\Delta_1 = 0$ and $\Delta_2 = \mu^* - \mu_2 = 0.3$, so that $\sum_j \Delta_j = 0.3$. Plugging these values in,

$$\mathbb{E}[R_{1000}] \leq \frac{8 \ln 1000}{0.3} + \left(1 + \frac{\pi^2}{3} \right) \cdot 0.3 \approx 185.49.$$

Therefore, on average, the UCB1 policy would yield at most about 185.5 fewer purchases than an oracle strategy that always displays the optimal advertisement. The regret upper bound of UCB1 is significantly lower than that of ETC, 286.25 (2).

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